

Logistic Regression Lab

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Q1

Variable pairs that are not correlated at all.

Price & Age

Price & Income

Q2

Are there any highly correlated variables in this dataset?

No there are no highly correlated variables (income and age are weakly correlated)

Q3

How many categories are there for the Price variable?

3 Categories

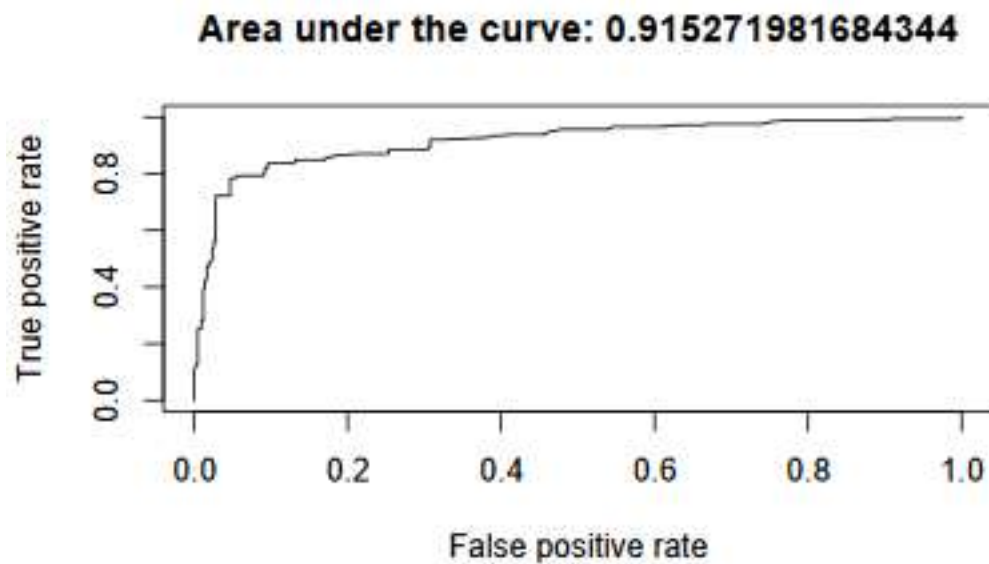
Q4

Why is it divided into two entries only in the model?

Because it uses dummy variables when there are n categories it makes n-1 variables

And we can get the third one from these two entries.

Q5



Q5.1 Write the value of this expression (just the number)

0.915272

Q5.2 What is the maximum value of AUC (ideal case)?

1

Q6

Classification Threshold

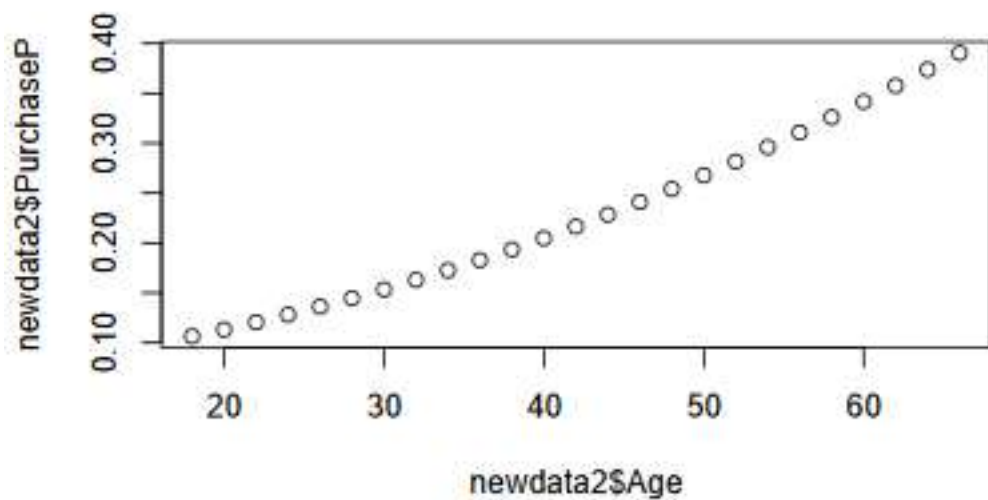
Q7

How is the predicted probability affected by changing only price holding all other variables constant?

When price increases probability decreases holding all other variables constant

Q8

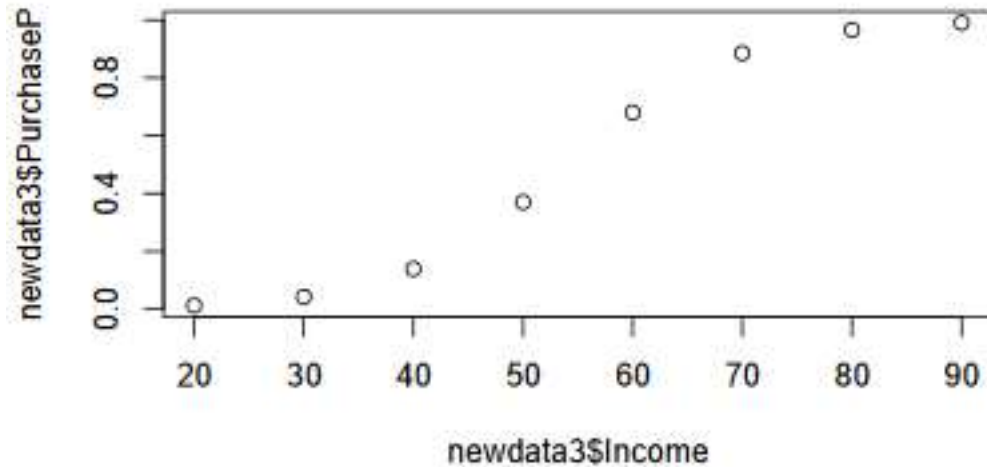
How is the predicted probability affected by changing only age holding all other variables constant?



When age increases probability increases (mostly linear relationship) holding all other variables constant

Q9

How is the predicted probability affected by changing only income holding all other variables constant?



When income increases probability increases (but not linear relationship) holding all other variables constant